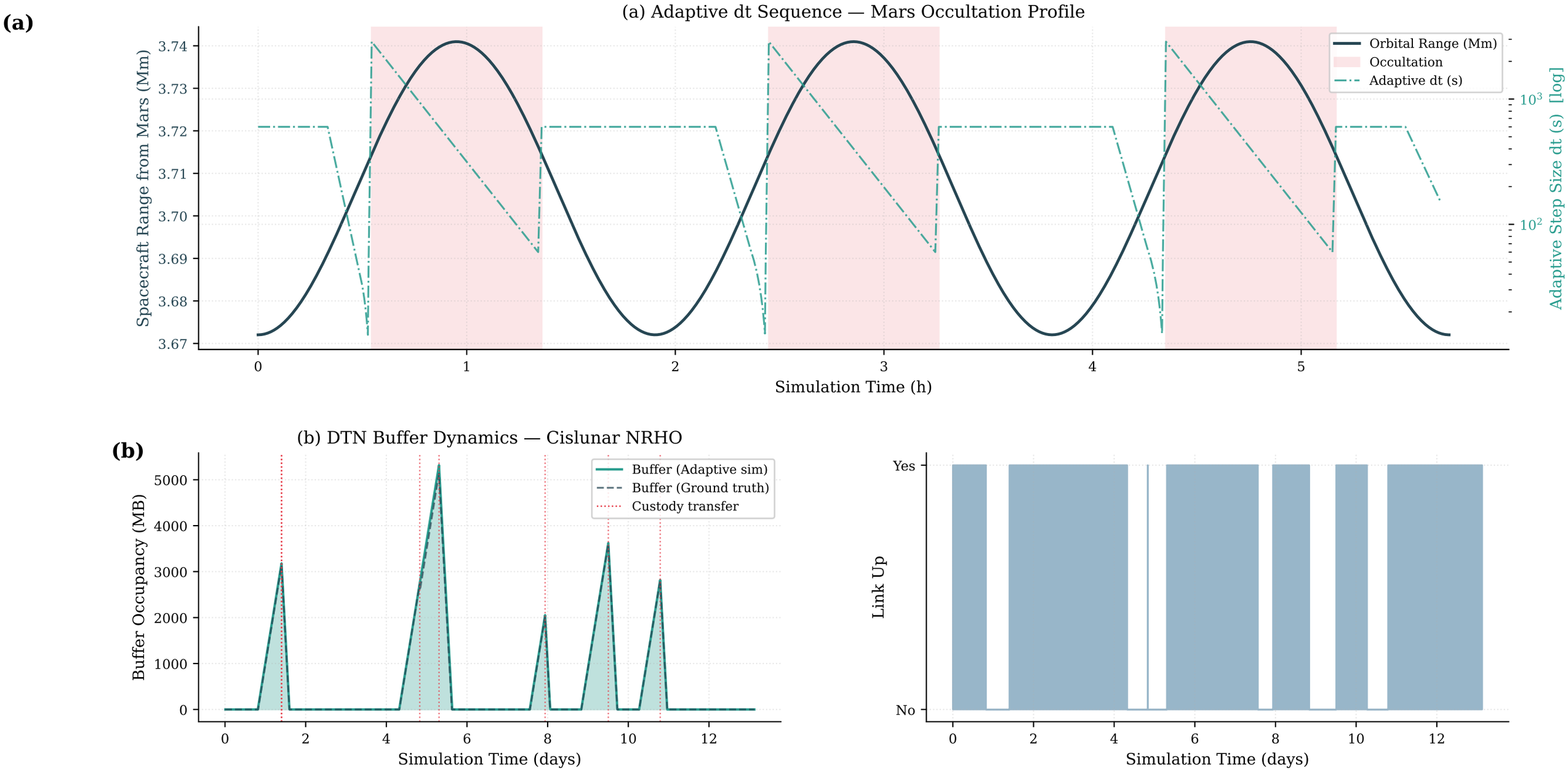

(a)
(a) Adaptive dt Sequence — Mars Occultation Profile
Spacecraft Range from Mars (Mm)
3.74
3.73
3.72
3.71
3.70
3.69
3.68
3.67
Orbital Range (Mm)
Occultation
Adaptive dt (s)
10^3
10^2
Adaptive Step Size dt (s) [log]
0
1
2
3
4
5
Simulation Time (h)
(b)
(b) DTN Buffer Dynamics — Cislunar NRHO
Buffer Occupancy (MB)
5000
4000
3000
2000
1000
0
Buffer (Adaptive sim)
Buffer (Ground truth)
Custody transfer
0
2
4
6
8
10
12
Simulation Time (days)
Link Up
Yes
No
0
2
4
6
8
10
12
Simulation Time (days)